

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The distance formula

1. The centre of a circle is $(x - 7, 2x)$. The circle passes through $(-9, 11)$ and its radius is $5\sqrt{2}$ units. Find the possible values of x . **CBSE 2026, 2 marks**

SOLUTION

The distance from the centre to $(-9, 11)$ equals the radius.

$$(x - 7 + 9)^2 + (2x - 11)^2 = (5\sqrt{2})^2$$

$$x^2 + 4x + 4 + 4x^2 - 44x + 121 = 50, \text{ so } 5x^2 - 40x + 75 = 0$$

$$x^2 - 8x + 15 = 0, \text{ so } (x - 3)(x - 5) = 0$$

Answer: $x = 3$ or $x = 5$

2. A circle has its centre at $(2a, a - 7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a . **CBSE 2025, 2 marks**

SOLUTION

Radius = $\frac{10\sqrt{2}}{2} = 5\sqrt{2}$ units, so the distance from the centre to $(11, -9)$ is $5\sqrt{2}$.

$$(2a - 11)^2 + (a - 7 + 9)^2 = (5\sqrt{2})^2$$

$$4a^2 - 44a + 121 + a^2 + 4a + 4 = 50, \text{ so } 5a^2 - 40a + 75 = 0$$

$$a^2 - 8a + 15 = 0, \text{ so } (a - 3)(a - 5) = 0$$

Answer: $a = 3$ or $a = 5$

3. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**

SOLUTION

D is the mid-point of BC : $D = \left(\frac{6+0}{2}, \frac{4+0}{2}\right) = (3, 2)$

$$AD = \sqrt{(5-3)^2 + (-6-2)^2} = \sqrt{4+64} = \sqrt{68}$$

Answer: (a)

4. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

SOLUTION

Both points have the same y -coordinate, so the distance is $|x - 3| = 15$.

$x - 3 = 15$ or $x - 3 = -15$, so $x = 18$ or $x = -12$.

Answer: (b)

5. Find the distance between $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$. (a) $2\sqrt{10}$ units (b) $4\sqrt{10}$ units (c) $2\sqrt{20}$ units (d) 0 **CBSE 2023, 1 mark**

SOLUTION

Distance = $\sqrt{(-2\sqrt{5} - 0)^2 + (0 - 2\sqrt{5})^2} = \sqrt{20 + 20} = \sqrt{40} = 2\sqrt{10}$ units

Answer: (a)

6. Two vertices of an equilateral triangle are $(-5, 3)$ and $(5, 3)$, and the origin lies inside the triangle. Find the third vertex. Take $\sqrt{3} = 1.7$. **CBSE 2023, 3 marks**

SOLUTION

The side joining the two given vertices has length 10, and its mid-point is $(0, 3)$.

The third vertex is equidistant from the two given points, so it lies on the y -axis. Let it be $(0, y)$.

$(0 - 5)^2 + (y - 3)^2 = 10^2$, so $(y - 3)^2 = 75$ and $y = 3 \pm 5\sqrt{3} = 3 \pm 8.5$

$y = 11.5$ puts the whole triangle above the line $y = 3$, so the origin would be outside it. Hence $y = -5.5$.

Answer: $(0, -5.5)$

7. Assertion (A): $P(0, 2)$ is the point where the line $3x + 2y = 4$ meets the y -axis. Reason (R): The distance of $P(0, 2)$ from the x -axis is 2 units. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2023, 1 mark**

SOLUTION

On the y -axis $x = 0$: $2y = 4$, so $y = 2$. The line meets the y -axis at $(0, 2)$, so A is true.

The distance of a point from the x -axis is its y -coordinate, here 2 units, so R is true. But R does not tell us why P lies on the line.

Answer: (b)

8. Find the point on the y -axis that is equidistant from $(5, -2)$ and $(-3, 2)$. **CBSE 2019, 3 marks**

SOLUTION

Let the point be $P(0, y)$. Equate the squares of the two distances.

$(0 - 5)^2 + (y + 2)^2 = (0 + 3)^2 + (y - 2)^2$

$25 + y^2 + 4y + 4 = 9 + y^2 - 4y + 4$, so $8y = -16$ and $y = -2$.

Answer: $(0, -2)$

9. Write the distance of the point $P(x, y)$ from the origin. **CBSE 2018, 1 mark**

SOLUTION

$$\text{Distance} = \sqrt{(x-0)^2 + (y-0)^2}$$

Answer: $\sqrt{x^2 + y^2}$ units

10. Show that $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle.

CBSE 2016, 2 marks

SOLUTION

Call the points $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$.

$$AB^2 = 3^2 + 4^2 = 25, AC^2 = (-4)^2 + 3^2 = 25 \text{ and } BC^2 = (-7)^2 + (-1)^2 = 50$$

$AB = AC$, so the triangle is isosceles. $AB^2 + AC^2 = 50 = BC^2$, so by the converse of the Pythagoras theorem $\angle A = 90^\circ$.

So the triangle is right-angled and isosceles. Proved.

11. The point $P(x, y)$ is equidistant from $A(a + b, b - a)$ and $B(a - b, a + b)$. Prove that $bx = ay$.

CBSE 2016, 3 marks

SOLUTION

$$PA^2 = PB^2: (x - a - b)^2 + (y - b + a)^2 = (x - a + b)^2 + (y - a - b)^2$$

$$\text{Use } p^2 - q^2 = (p - q)(p + q) \text{ on the } x \text{ terms: } (x - a - b)^2 - (x - a + b)^2 = (-2b)(2x - 2a) = -4bx + 4ab$$

$$\text{On the } y \text{ terms: } (y - b + a)^2 - (y - a - b)^2 = (2a)(2y - 2b) = 4ay - 4ab$$

$$\text{Adding the two results gives } 0: -4bx + 4ab + 4ay - 4ab = 0, \text{ so } 4ay = 4bx.$$

So $bx = ay$. Proved.

2 The section formula

12. Where does the mid-point of the segment joining $(5, -4)$ and $(6, 4)$ lie? (a) on the x -axis (b) on the y -axis (c) at the origin (d) on neither axis **CBSE 2026, 1 mark**

SOLUTION

$$\text{Mid-point} = \left(\frac{5+6}{2}, \frac{-4+4}{2} \right) = \left(\frac{11}{2}, 0 \right)$$

Its y -coordinate is 0 and its x -coordinate is not 0.

Answer: (a)

- 13.** In what ratio does the x -axis divide the segment joining $(-6, 5)$ and $(-4, -1)$? Find the point where they meet. **CBSE 2026, 3 marks**

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-4k - 6}{k + 1}, \frac{-k + 5}{k + 1}\right)$.

On the x -axis the y -coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$x = \frac{-4(5) - 6}{5 + 1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio 5 : 1; the point is $\left(-\frac{13}{3}, 0\right)$

- 14.** Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? **CBSE 2025, 1 mark**

SOLUTION

$$\text{Mid-point} = \left(\frac{-4 + 4}{2}, \frac{5 + 6}{2}\right) = \left(0, \frac{11}{2}\right)$$

Its x -coordinate is 0.

Answer: on the y -axis

- 15.** In what ratio does the y -axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. **CBSE 2025, 3 marks**

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-k + 5}{k + 1}, \frac{-4k - 6}{k + 1}\right)$.

On the y -axis the x -coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$y = \frac{-4(5) - 6}{5 + 1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio 5 : 1; the point is $\left(0, -\frac{13}{3}\right)$

- 16.** A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**

SOLUTION

The centre is the mid-point of the diameter. Let the other end be (x, y) .

$$\frac{6 + x}{2} = 2 \text{ gives } x = -2, \text{ and } \frac{0 + y}{2} = 0 \text{ gives } y = 0.$$

Answer: (c)

- 17.** The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment?
Find y as well. **CBSE 2024, 3 marks**

SOLUTION

Let the ratio be $k : 1$. Then $\frac{2k+1}{k+1} = \frac{8}{5}$.

$10k + 5 = 8k + 8$, so $k = \frac{3}{2}$. The ratio is $3 : 2$.

$$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$$

Answer: ratio $3 : 2$ and $y = \frac{13}{5}$

- 18.** $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other.

CBSE 2024, 3 marks

SOLUTION

$$P = \left(-1, \frac{5}{2}\right), Q = (1, 6), R = \left(3, \frac{5}{2}\right), S = (1, -1)$$

$$\text{Mid-point of diagonal } PR = \left(\frac{-1+3}{2}, \frac{5/2+5/2}{2}\right) = \left(1, \frac{5}{2}\right)$$

$$\text{Mid-point of diagonal } QS = \left(\frac{1+1}{2}, \frac{6-1}{2}\right) = \left(1, \frac{5}{2}\right)$$

The diagonals PR and QS have the same mid-point, so they bisect each other.

- 19.** In what ratio does the x -axis divide the segment joining $A(3, 6)$ and $B(-12, -3)$? (a) $1 : 2$ (b) $1 : 4$ (c) $4 : 1$ (d) $2 : 1$ **CBSE 2023, 1 mark**

SOLUTION

Let the ratio be $k : 1$. On the x -axis the y -coordinate is 0: $\frac{-3k+6}{k+1} = 0$, so $k = 2$.

Answer: (d)

- 20.** AB is a diameter of a circle with centre $(2, -3)$, and B is the point $(1, 4)$. Find the coordinates of A .

CBSE 2019, 1 mark

SOLUTION

The centre is the mid-point of the diameter. Let $A = (x, y)$.

$$\frac{x+1}{2} = 2 \text{ gives } x = 3, \text{ and } \frac{y+4}{2} = -3 \text{ gives } y = -10.$$

Answer: $A(3, -10)$

- 21.** In what ratio does the x -axis divide the segment joining $(1, -3)$ and $(4, 5)$? Find the point of division.

CBSE 2019, 2 marks

SOLUTION

Let the ratio be $k : 1$. On the x -axis the y -coordinate is 0: $\frac{5k - 3}{k + 1} = 0$, so $k = \frac{3}{5}$.

The ratio is $3 : 5$. Then $x = \frac{3 \times 4 + 5 \times 1}{3 + 5} = \frac{17}{8}$.

Answer: ratio $3 : 5$; the point is $\left(\frac{17}{8}, 0\right)$

- 22.** The segment joining $A(2, 1)$ and $B(5, -8)$ is trisected at P and Q , with P nearer to A . P also lies on the line $2x - y + k = 0$. Find k . CBSE 2019, 3 marks

SOLUTION

P divides AB in the ratio $1 : 2$.

$$P = \left(\frac{1 \times 5 + 2 \times 2}{3}, \frac{1 \times (-8) + 2 \times 1}{3} \right) = (3, -2)$$

P lies on the line: $2(3) - (-2) + k = 0$, so $8 + k = 0$.

Answer: $k = -8$

- 23.** In what ratio does $P(4, m)$ divide the segment joining $A(2, 3)$ and $B(6, -3)$? Find m . CBSE 2018, 2 marks

SOLUTION

Let the ratio be $k : 1$. Compare x -coordinates: $\frac{6k + 2}{k + 1} = 4$, so $6k + 2 = 4k + 4$ and $k = 1$.

The ratio is $1 : 1$, so P is the mid-point: $m = \frac{3 + (-3)}{2} = 0$.

Answer: ratio $1 : 1$; $m = 0$

- 24.** $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$. Find a and b , and the lengths of its sides. CBSE 2018, 3 marks

SOLUTION

The diagonals of a parallelogram bisect each other, so AC and BD have the same mid-point.

$$\frac{-2 + 4}{2} = \frac{a + 1}{2} \text{ gives } a = 1, \text{ and } \frac{1 + b}{2} = \frac{0 + 2}{2} \text{ gives } b = 1.$$

$$AB = \sqrt{(1 + 2)^2 + (0 - 1)^2} = \sqrt{10} \text{ and } BC = \sqrt{(4 - 1)^2 + (1 - 0)^2} = \sqrt{10}$$

Opposite sides are equal, so $CD = \sqrt{10}$ and $DA = \sqrt{10}$.

Answer: $a = 1$, $b = 1$; each side is $\sqrt{10}$ units

25. P and Q are the points of trisection of the segment joining $A(2, -2)$ and $B(-7, 4)$, with P nearer to A . Find P and Q . **CBSE 2016, 2 marks**

SOLUTION

$$P \text{ divides } AB \text{ in the ratio } 1 : 2: P = \left(\frac{-7 + 4}{3}, \frac{4 - 2}{3} \right) = (-1, 0)$$

$$Q \text{ divides } AB \text{ in the ratio } 2 : 1: Q = \left(\frac{-14 + 2}{3}, \frac{8 - 2}{3} \right) = (-4, 2)$$

Answer: $P(-1, 0)$ and $Q(-4, 2)$

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